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C:\WINDOWS\system32\cmd. X + v
Requirement already satisfied: annotated-types>=0.6.0 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain-classic<2.0.0,>=1.0.7->langchain-community) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.4 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain-classic<2.0.0,>=1.0.7->langchain-community) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain-classic<2.0.0,>=1.0.7->langchain-community) (0.4.2)
Requirement already satisfied: python-dotenv>=0.21.0 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic-settings<3.0.0,>=2.10.1->langchain-community) (1.2.2)
Requirement already satisfied: charset-normalizer<4,>=2 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests<3.0.0,>=2.32.5->langchain-community) (3.4.9)
Requirement already satisfied: urllib3<3,>=1.26 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests<3.0.0,>=2.32.5->langchain-community) (2.7.0)
Requirement already satisfied: greenlet>=1 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy<3.0.0,>=1.4.0->langchain-community) (3.5.5)
Downloading langchain_community-0.4.2-py3-none-any.whl (2.4 MB)
2.4/2.4 MB 1.4 MB/s 0:00:01
Downloading httpx_sse-0.4.3-py3-none-any.whl (9.0 kB)
Installing collected packages: httpx-sse, langchain-community
Successfully installed httpx-sse-0.4.3 langchain-community-0.4.2

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\Sanjana CA>pip show langchain-community
Name: langchain-community
Version: 0.4.2
Summary: Community contributed LangChain integrations.
Home-page: https://docs.langchain.com/
Author:
Author-email:
License: MIT
Location: C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\Lib\site-packages
Requires: aiohttp, httpx-sse, langchain-classic, langchain-core, langsmith, numpy, pydantic-settings, pyyaml, requests, sqlalchemy, tenacity
Required-by:

C:\Users\Sanjana CA>
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ore==1.*->httpx<1,>=0.23.0->langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.21->langchain-ollama) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain-core<2.0.0,>=1.2.21->langchain-ollama) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.4 in C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain-core<2.0.0,>=1.2.21->langchain-ollama) (2.46.4)
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[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D2_LangChain_Memory> python w6d2_langchain_memory.py

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TASK 1: LANGCHAIN CHAIN
PromptTemplate -> Ollama LLM -> OutputParser
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Input 1: What is artificial intelligence?
Output 1: Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and act like humans. It involves developing algorithms and software that can learn, make decisions, understand natural language, recognize patterns, solve problems, and perform other tasks typically requiring human cognition. AI systems can be categorized into narrow or weak AI (which is designed for specific tasks) and general or strong AI (which would possess the capability to understand and think like a human across all domains).

Input 2: What is machine learning?
Output 2: Machine learning is a subset of artificial intelligence that enables systems to improve their performance in a specific task through experience without being explicitly programmed. It involves training algorithms using large datasets so they can make predictions or decisions based on data patterns, improving accuracy over time.

Input 3: What is LangChain?
Output 3: LangChain is a collection of tools designed to help with common tasks in Natural Language Processing (NLP), including text processing, summarization, chatbot creation, etc. It provides various tools written in Python for developers to build applications more efficiently. LangChain aims to simplify and standardize the process of working with natural language data.
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re efficiently. LangChain aims to simplify and standardize the process of working with natural language data.

Input 4: What is an API?
Output 4: An API (Application Programming Interface) is a set of rules and protocols for exchanging data between software components. It defines how different software programs can communicate with each other, enabling them to interact and share data securely. APIs are used in various applications like mobile apps, web services, and software development to allow seamless integration and collaboration among different systems.

Input 5: What is Python?
Output 5: Python is an interpreted high-level programming language known for its readability and simplicity. It supports multiple programming paradigms and has a large standard library, making it suitable for web development, scientific computing, data analysis, artificial intelligence, automation, and more. Python is open-source and has a vibrant community that contributes to its continued growth and improvement.

TASK 1 COMPLETED: 5 inputs tested.

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TASK 2: CONVERSATION HISTORY
5 conversation turns
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Turn 1
User: My name is Alex.
Assistant: Hello Alex! It's nice to meet you. How can I assist you today? Do you have any specific questions or topics that you'd like to discuss?

Turn 2
User: I am learning Python.
Assistant: Great to hear that Alex! Learning Python is a fantastic step towards becoming proficient in software development. What are you currently focusing on? Are there any specific projects or concepts that you're having trouble with, or perhaps you'd like some general advice on how to improve your skills?

Turn 3
User: What programming language am I learning?
Assistant: You are learning Python. Is there anything specific about Python that you would like to know more about or any challenges you're currently facing in your studies?
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Turn 4
User: What is my name?
Assistant: Your name is Alex. How can I assist you further with your Python learning?

Turn 5
User: What am I currently learning?
Assistant: It seems like there might be a bit of confusion here, Alex. To clarify: You are currently learning Python. Is there anything specific about Python you'd like to know more about or any challenges you're facing in your studies?

Stored Conversation History:
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Turn 1
User: My name is Alex.
Assistant: Hello Alex! It's nice to meet you. How can I assist you today? Do you have any specific questions or topics that you'd like to discuss?
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Turn 2
User: I am learning Python.
Assistant: Great to hear that Alex! Learning Python is a fantastic step towards becoming proficient in software development. What are you currently focusing on? Are there any specific projects or concepts that you're having trouble with, or perhaps you'd like some general advice on how to improve your skills?
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Turn 3
User: What programming language am I learning?
Assistant: You are learning Python. Is there anything specific about Python that you would like to know more about or any challenges you're currently facing in your studies?
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Turn 4
User: What is my name?
Assistant: Your name is Alex. How can I assist you further with your Python learning?
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Turn 5
User: What am I currently learning?
Assistant: It seems like there might be a bit of confusion here, Alex. To clarify: You are currently learning Python. Is there anything specific about Python you'd like to know more about or any challenges you're facing in your studies?
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TASK 2 COMPLETED: Conversation history maintained across 5 turns.
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TASK 3: SIMPLE LANGCHAIN AGENT
Tools: Web Search Stub + Calculator
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Agent Task 1
User: Calculate 25 * 4 + 10.
Agent: The result of the calculation 25 * 4 + 10 is 110.

Agent Task 2
User: Use the web search stub to search for information about LangChain.
Agent: The web search stub found that LangChain is a framework designed to build applications using language models, along with various components like prompts, memory, and more. It connects these elements for enhanced functionality. Do you have any specific questions about LangChain or need further information?

Agent Task 3
User: Calculate 144 / 12 and explain the result briefly.
Agent: The calculation of 144 divided by 12 equals 12.0. This means that 12 goes into 144 exactly 12 times.

TASK 3 COMPLETED: Agent tested with 3 tasks.

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W6D2 PRACTICAL TASK SUMMARY
=====
Task 1 - LangChain Chain: COMPLETED
5 inputs tested

Task 2 - Conversation History: COMPLETED
5 conversation turns tested

Task 3 - LangChain Agent: COMPLETED
2 tools configured
3 agent tasks tested
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Agent Task 1
User: Calculate 25 * 4 + 10.
Agent: The result of the calculation 25 * 4 + 10 is 110.

Agent Task 2
User: Use the web search stub to search for information about LangChain.
Agent: The web search stub found that LangChain is a framework designed to build applications using language models, along with various components like prompts, memory, and more. It connects these elements for enhanced functionality. Do you have any specific questions about LangChain or need further information?

Agent Task 3
User: Calculate 144 / 12 and explain the result briefly.
Agent: The calculation of 144 divided by 12 equals 12.0. This means that 12 goes into 144 exactly 12 times.

TASK 3 COMPLETED: Agent tested with 3 tasks.

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W6D2 PRACTICAL TASK SUMMARY
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Task 1 - LangChain Chain: COMPLETED
5 inputs tested

Task 2 - Conversation History: COMPLETED
5 conversation turns tested

Task 3 - LangChain Agent: COMPLETED
2 tools configured
3 agent tasks tested

Tools used:
1. Web Search Stub
2. Calculator

W6D2 PRACTICAL IMPLEMENTATION FINISHED
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D2_LangChain_Memory>
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